claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a120c362eb761312f.jsonl`
- SHA-256: `c3622b4430ba37f5b00d3f53f4994bf9d5f3499e9ef2c46689038144f6486ff8`
- Source modified: `2026-06-06T17:31:52+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf_6237b110-d58`
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`

Transcript

1. user (2026-06-06T17:31:01.361Z)

Source Extractor

```
Research question: "Research alternative methods to automatically detect rally start/end
segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-
HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation
is the best approach or whether other methods are competitive or complementary. Keep it focused
("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches
over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):
- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY
inference at runtime ? privacy lever: user video never leaves the device. It's a commercial
product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-
commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We
derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We
ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision
ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label
rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features).
Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the
learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck
(its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes /
pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd
noise, umpire/score calls).

SPECIFICALLY COVER:
1. Method families for rally/point segmentation in racket sports (badminton, tennis, table
tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep
models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance"
gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-
overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM
teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference
COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their
licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher
? local-distillation the best, or are some alternatives competitive or complementary ?
especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run
fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory +
person detector + audio)."
```

Fetch and extract key claims from this source:
URL: https://arxiv.org/html/2409.11760v1
Title: Sound-Based Spin Estimation in Table Tennis: Dataset and Real-Time Classification Pipeline (Sony AI)
Found via: practitioner/local-levers search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
   - be a concrete, checkable statement (not vague generalities)
   - include a direct quote from the source as support
   - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-06T17:31:05.448Z)

I'll fetch the source and extract key claims relevant to the research question.

3. user (2026-06-06T17:31:24.381Z)

Key Claims Extraction: Spin Detection Using Racket Bounce Sounds in Table Tennis

Publication Details:
- Date: September 16, 2024
- Authors: Thomas Gossard, Julian Schmalzl, Andreas Ziegler, Andreas Zell
- Institution: Cognitive Systems Group, University of Tuebingen, Germany
- Funding: Sony AI

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(1) Sound-Based Detection of Racket-Sport Events

Table Tennis Ball Hits & Spin Detection:
- "The sound generated when the ball strikes the racket can assist in predicting the ball's trajectory, especially in determining the spin."
- "We show that the sound generated by the racket can indicate whether spin was applied to the ball, or not."
- "Incorporating auditory data has been shown to improve the prediction of ball spin type in table tennis."

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(2) Dataset Collected

Size & Composition:
- "We created a comprehensive dataset featuring bounce sounds from 10 racket configurations, each applying various spins to the ball."
- Total samples: 3,396 racket bounces + 2,296 non-bounce samples (table, floor, other objects)
- "The audio samples with spin applied on the ball were recorded for serves."
- Recording equipment: "Zoom H4n Pro Handy Recorder, a directional microphone, at 44.1kHz"

License/Availability:
- "Our comprehensive racket-ball bounce sound dataset is available at: https://github.com/cogsys-tuebingen/tt_sounds"
- (No explicit license statement provided in the document)

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(3) Classification Accuracy & F1 Scores

CNN Performance (Best Method):

Task	Precision	Recall	F1
Spin	0.98	0.95	0.96
Surface	0.97	0.97	0.97

Comparison with Other Methods:
- SVM: Spin F1=0.93, Surface F1=0.94
- GMM: Spin F1=0.17, Surface F1=0.06

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(4) Real-Time & Local Compute Requirements

Latency:
- "The detection algorithm shows high temporal resolution, detecting energy peaks on average no later than 0.2 ms after occurring."
- "This allows for energy peak detection with minimal latency, as sudden energy peaks generally get detected within a millisecond of their onset."

Computational Approach:
- "We adopt a two-stage approach...similar to previous work: First, we use energy-based peak detection to identify potential bounce sounds. Then, a CNN classifies the detected peaks."
- No explicit compute requirements (CPU/GPU specifications) mentioned.

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(5) Audio Features & Methods

Microphone Setup:
- "The microphone was always oriented towards the player at distances ranging from 50 cm to 2 m."

Feature Extraction:
- "The audio is...transformed into a Mel spectrogram."
- "Experimental results demonstrated that the CNN performed best using 64 Mel bands and an FFT window length of 256 samples."
- "Mel spectrogram consistently produced the best results" (compared to raw spectrograms, MFCCs, LFCCs)

Neural Network Architecture:
- "We utilized a six-layer CNN for bounce sound classification, with each layer consisting of a 2D convolution, ReLU activation, and batch normalization."
- "Categorical cross-entropy was used as the loss function."

Classical Methods Tested:
- GMM (Gaussian Mixture Model) with diagonal covariance
- SVM with linear kernel
- Both outperformed by CNN

Signal Processing:
- "The audio signal is first filtered using a 5th-order Butterworth high-pass filter with a cutoff frequency of 10 kHz."
- "We apply zero-phase filtering to ensure high temporal accuracy."

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(6) Ball-Bat Contact/Hit Timing Detection

Temporal Precision:
- "To achieve millisecond-level temporal accuracy, we first detect high-frequency peaks that may correspond to table tennis ball bounces."
- Peak detection onset accuracy: 0.09 ms (no noise), 0.2 ms (with noise)
- Audio window: "standardizing the length to 661 samples per data point, corresponding to 15 ms of audio"

Contact Duration Reference:
- "When the ball bounces against a table tennis racket, the contact duration ranges from 1.3 ms to 1.8 ms"

4. assistant (2026-06-06T17:31:30.075Z)

I have comprehensive extraction from this primary research source. Let me assess and structure the findings relevant to the badminton rally-segmentation research question.

This is a primary peer-reviewed/arXiv research paper from University of Tübingen, funded by Sony AI. The relevance to the research question is via the AUDIO-based hit detection family ? table tennis (a racket sport) using sound to detect ball-racket contact with millisecond timing, which is the most transferable finding for the local-first audio approach.

5. user (2026-06-06T17:31:41.193Z)

Structured output provided successfully

6. assistant (2026-06-06T17:31:52.247Z)

Extraction complete. Summary of what I found in this source:

Source: https://arxiv.org/html/2409.11760v1 ? "Sound-Based Spin Estimation in Table Tennis" (Gossard, Schmalzl, Ziegler, Zell; University of Tübingen, funded by Sony AI; published 2024-09-16).

Source quality: primary (arXiv research paper from a university lab).

Relevance to the badminton rally-segmentation research question: The paper is about table tennis spin estimation, not rally segmentation, but it is directly informative for the **audio-based hit-detection family** that is one of the locally-runnable levers in our context. Key transferable takeaways:

1. A **two-stage audio pipeline** ? cheap energy-based peak detection (no ML) to localize contact events, then a small CNN to classify ? detects racket-ball contact with sub-millisecond-to-0.2ms temporal precision even under noise. This is exactly the "thwack" hit-detection primitive we'd want for deriving rally boundaries locally.
2. A **six-layer CNN over Mel spectrograms** hits F1 0.96-0.97, vastly beating SVM/GMM ? so a tiny, CPU/edge-runnable model suffices once peaks are isolated.
3. Cheap, portable DSP front-end: 5th-order Butterworth high-pass at 10 kHz to isolate the high-frequency contact transient, 15 ms windows.
4. An **open dataset** (3,396 bounce + 2,296 non-bounce samples) is released at github.com/cogsys-tuebingen/tt_sounds, but the page states **no explicit license** ? a commercial-viability caveat to flag.

Caveat for our use: this is table-tennis, indoor, close-mic'd (50 cm-2 m, directional mic at 44.1 kHz) ? real-world badminton broadcast/handheld audio with crowd noise and farther mics would be a harder regime, so the F1 numbers are an upper bound for clean close-mic conditions, not broadcast.